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Lab 1: Requirements Table

Problem Statement #27 — Warehouse Inventory & Pallet Location Tracker | Actors: Warehouse Operator, Logistics Supervisor

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage.	High	Pass: Pallet assigned to valid rack. Fail: Placement permitted on rack where total weight exceeds rated structural limit.	Prevents structural failure/collapse from overloaded racks.
FR-002	Functional	The system shall allow a Warehouse Operator to log a stock movement (inbound/outbound) by scanning a pallet's barcode or RFID tag.	High	Pass: Scan updates pallet status and timestamp in the movement log within 2 seconds. Fail: Scan is accepted but no log entry is created.	Ensures traceability of every physical stock movement.
FR-003	Functional	The system shall allow a Logistics Supervisor to search for a pallet's current shelf location by SKU or pallet ID.	Medium	Pass: Query returns exact bin coordinates for a valid SKU. Fail: Query returns no result or an incorrect bin for a SKU known to be in stock.	Enables quick retrieval and audit without manual shelf search.
FR-004	Functional	The system shall allow a Warehouse Operator to relocate a pallet from one shelf position to another, revalidating the destination rack's weight capacity before confirming the move.	High	Pass: Relocation confirmed only if destination rack remains within rated limit. Fail: Relocation confirmed despite destination exceeding capacity.	Prevents overload during reslotting/relocation operations.
FR-005	Functional	The system shall generate a capacity alert to the Logistics Supervisor when a rack's total load reaches 90% of its rated maximum.	Medium	Pass: Alert is raised and visible on the supervisor dashboard within 5 seconds of the threshold being crossed. Fail: Rack exceeds 90% with no alert generated.	Provides early warning before a rack reaches its hard limit.
NFR-001	Nonfunctional (Performance & Security)	The warehouse item lookup service must locate any SKU pallet coordinate across 100,000 bins in under 100 ms.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures responsive lookups even at full warehouse scale.
NFR-002	Nonfunctional (Reliability / Auditability)	The system shall retain a complete, immutable audit log of all stock movement and placement events for a minimum of 12 months.	Medium	Pass: Any movement/placement event from the past 12 months can be retrieved with operator ID, timestamp, and pallet ID intact. Fail: Any such event is missing, altered, or unretrievable.	Supports compliance, dispute resolution, and inventory audits.

Use-Case Flow Specification

Problem Statement #27 — Warehouse Inventory & Pallet Location Tracker

Use Case: UC-03 Place Pallet

Primary Actor: Warehouse Operator

Supporting Actors: RFID/Barcode Scanner (device), Logistics Supervisor (notified on alert) **Preconditions:**

1. The Warehouse Operator is authenticated and logged into the system.
2. The pallet to be stored has a valid, registered barcode/RFID tag with an associated SKU.
3. The warehouse rack layout and current load data for all racks are available to the system.

Postconditions:

1. The pallet's 3D shelf coordinates are recorded and associated with its SKU in the system.
2. The destination rack's current load total is updated to include the new pallet's weight.
3. A stock movement ("inbound / placement") entry is appended to the audit log.

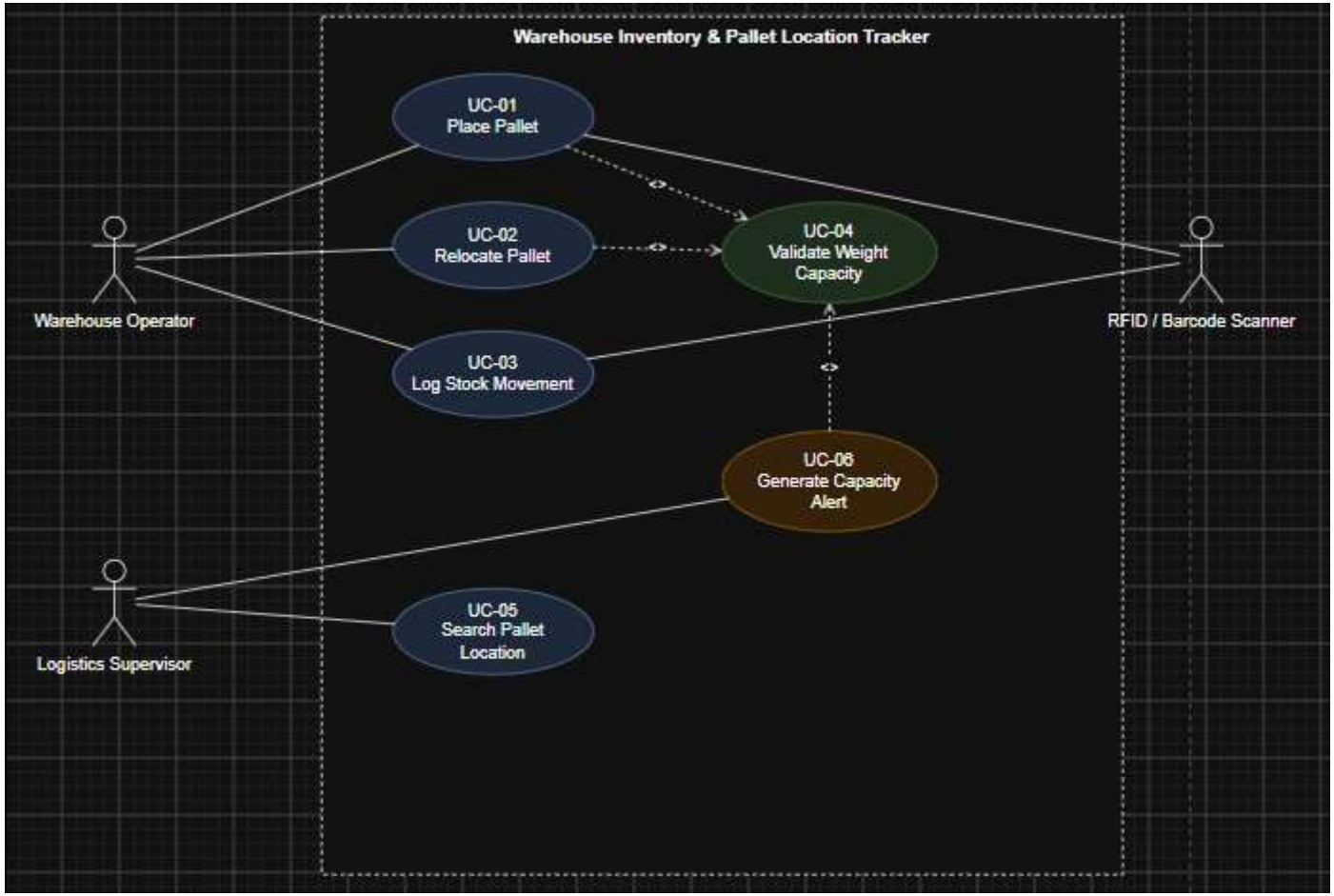
Main Success Scenario:

1. Warehouse Operator selects "Place Pallet" and scans the pallet's barcode/RFID tag.
2. System retrieves the pallet's SKU, weight, and dimensions from the scan.
3. System proposes an available shelf/bin coordinate for the pallet.
4. Operator confirms (or manually overrides) the proposed coordinates.
5. System validates the proposed placement against the destination rack's maximum load capacity (includes UC: Validate Weight Capacity).
6. System confirms the placement, updates the rack's total load, and assigns the pallet to the bin.
7. System logs the placement event (pallet ID, bin coordinates, operator, timestamp) to the audit log.
8. System displays a confirmation message with the final bin location to the Operator.
9. Use case ends successfully.

Alternate Flow:

5a. Rack Capacity Exceeded

a) System determines the proposed rack would exceed its rated maximum load if the pallet is placed there.
b) System rejects the proposed placement and displays a "Capacity Exceeded" message to the Operator.
c) System suggests the next available rack/bin with sufficient remaining capacity.
d) Operator accepts the suggested bin (flow resumes at main step 6) or cancels the placement, ending the use case without storing the pallet.


Warehouse Inventory & Pallet Location Tracker
UC-01
Place Pallet
UC-02
Relocate Pallet
UC-03
Log Stock Movement
UC-04
Validate Weight
Capacity
UC-05
Search Pallet
Location
UC-06
Generate Capacity
Alert
Warehouse Operator
Logistics Supervisor
RFID / Barcode Scanner